

XINGLONG(ALEX) SUN

Tel: 217-721-4650

www.xinglongsun.com linkedin.com/in/xinglong-alex-sun-4584b2174/ alexsxl417@gmail.com

EDUCATION

Stanford University

Aug. 2022 – May 2024

Master of Science in Computer Science

GPA: 4.0/4.0

University of Illinois at Urbana-Champaign (UIUC)

Aug. 2018 – Dec 2021

Bachelor of Science in Computer Engineering

(Rank No.1 of Class) GPA: 4.0/4.0

- **Honor:** Bronze Tablet Graduation Award, Highest Graduation Honor, Henry O. Koehler Merit Scholarship, Tan Family Education Foundation Scholarship, 2020-2021 Oakley Award in Electrical and Computer Engineering, Frank C. Mock Scholarship, Yunni and Maxine Pao Memorial Scholarship, 2019-2020 Oakley Award in Electrical and Computer Engineering, James Scholar, Dean's List, IEEE Member, HKN Member
- **Academic Services:** Reviewer for CVPR, TPAMI, RAL, IROS, and WACV; TA for CS103, CS330 at Stanford

RESEARCH EXPERIENCE AND AFFILIATIONS

NVIDIA

May 2024 – Present

Research Scientist @ Spatial Intelligence Lab

Santa Clara, CA

- Conduct research in Physical AI and world modeling, focusing on human-aligned evaluation and reward modeling, VLM judge, and post-training pipeline.
- Contributed to research that achieved first place in the CVPR 2025 End-to-End Driving Challenge.
- Contributed to **Cosmos-Transfer-1**, a world-to-world generative transfer model.
- Authored and contributed to multiple projects published at top conferences, e.g. CVPR, AAAI, IROS, and ICRA.

NVIDIA

Mar. 2022 – Dec. 2022; Jun. 2023 - Sept.2023

Research Intern @ AV Applied Research Lab

Santa Clara, CA

- Worked on efficient vision and model acceleration.
- Proposed a novel hardware-aware pruning framework generalizing across transformers and CNNs. **CVPR 2025**
- Proposed a novel dynamic sparse training algorithm inspired by exploitation-exploration. **WACV 2025**
- Helped to effectively apply latency-targeted pruning on detection models for realtime perception. **IEEE IV 2023**

Stanford University

Dec. 2022 – Present

Researcher @ Geometric Computation Lab with Prof. Leo Guibas

Stanford, CA

- Worked on visual tracking and video understanding.
- Proposed a novel algorithm on refining motion model in a self-supervised manner. **ICAR 2024**
- Contributed to an efficient dense point tracking framework for high-resolution imagery. **ICCV 2025**

INRIA and UIUC

July. 2021 – Sept. 2022

Researcher @ WILLOW Group of INRIA @ Computer Vision Group of UIUC

Urbana, IL

- Worked on depth completion using LIDAR point clouds.
- Proposed a novel depth completion model leveraging deformable convolution. **IROS 2023**

GaTech and UIUC

Dec. 2020 – Present

Researcher @ SHI/IFP Group with Prof. Humphrey Shi

Urbana, IL

- Worked on efficient deep learning and model compression.
- Proposed a novel sparsification algorithm for multitask networks. **CVPR 2022**
- Designed a novel structured filter pruning saliency for convolution neural networks. **WACV 2024**

SELECTED PUBLICATIONS

As of July, 2026: **14** conference publications (**7** first-authored) and **7** preprints on arXiv. Selected works below.

1. MDP: Multidimensional Vision Model Pruning with Latency Constraint
X. Sun, B Lakshmanan, M Shen, S Lan, J Chen, J Alvarez
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
2. AllTracker: Efficient Dense Point Tracking at High Resolution
A Harley, Y You, **X. Sun**, Y Zheng, N Raghuraman, Y Gu, S Liang, W Chu, A Dave, P Tokmakov, S You, R Ambrus, K Fragkiadaki, L Guibas
IEEE/CVF International Conference on Computer Vision (ICCV), 2025
3. Generalized trajectory scoring for end-to-end multimodal planning
Z. Li, W. Yao, Z. Wang, **X. Sun**, J. Chen, N. Chang, M. Shen, Z. Wu, S. Lan, and J. M. Alvarez
CVPR 2025 E2E Driving Global Challenge Winner
4. HAD: Combining Hierarchical Diffusion with Metric-Decoupled RL for End-to-End Driving
W. Yao, **X. Sun**, Z. Li, S. Lan, Z. Wang, J. M. Alvarez, and Z. Wu
IEEE/CVF European Conference on Computer Vision (ECCV), 2026
5. Cosmos-Transfer1: Conditional World Generation with Adaptive Multimodal Control
NVIDIA (**X. Sun** as contributor)
Arxiv White Paper
6. DriveJudge: Rethinking Autonomous Driving Evaluation with Vision-Language Models
X. Sun, K. Xie, J. Schmalfluss, D. Paschalidou, X. Zhang, S. Fidler, K. Chitta, and J. M. Alvarez
Arxiv Paper

INDUSTRIAL EXPERIENCES

NVIDIA

May 2021 - Sept. 2021

Architecture Energy Modelling Intern

Santa Clara, CA

- Developed an energy modeling tool that utilizes the GPU perf. data to estimate power consumption. Built a robust data mining and processing flow to handle inputs of many formats. Multi-processed to scale for large datasets.
- Developed a Python package to perform what/if analysis that reflects power consumption changes as high-level architectural modifications are applied. Supported correlation studies from multiple platforms including RTL, simulation, emulation, and silicon. Used data visualization to improve architects' debugging experience.
- Utilized clustering algorithm to compress and obtain more interpretable power analysis

Apple

Jun. 2020 - Sept. 2020

Hardware Tech Intern at SEG-CPU Team

Austin, TX

- Designed and implemented a set of automation tools in Python for CPU microarchitecture analysis and performance verification. Examples include a customized automated email notification system that alerts users of immediate major CPU performance issues and a large drift of some counter values through data harvesting and analysis.
- Devised a set of visualization tools in Python. Examples include Flask-based dashboard websites which visualize the drift of CPU IPC in dataplots and the "event" coverages in datatables. The dashboard supports several features like filtering, dynamic sorting, click&&hover effects by callbacks, etc.
- Designed and programmed a GUI using Kivy in Python for users to visualize the CPU pipeline. Code is multi-processed to optimize the performance. Implemented KIVY RecyclerView to scale when data's large.
- Created and maintained several SQL databases. Developed related codes to check and update the dbs periodically.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Assembly, SystemVerilog, JavaScript, HTML, CSS, SQL, Swift, Matlab

Tools: Claude, Codex, Cursor, PyTorch, Tensorflow, Git, SQL, Scikit-learn, ROS, Jupyter, LaTeX, AWS, Docker

Languages: English, Chinese, French